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Study on the Performance of Wi-Fi 6 Networks through MATLAB Multinode Simulation

Communications Systems Project

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1.

1. Abstract 3
2. Current status of implementation for the issue studied 4
2.1 Current implementation in MATLAB/ WLAN Toolbox 4
2.2 Limitations and design decisions 4
3. Theoretical basis 5
3.1 WLAN network architecture and the purpose of simulation 5
3.2 The MAC in IEEE 802.11 5
3.3 The PHY and SINR mapping→ PER 6
3.4 Propagation models and Tgax channels 6
3.5 Performance metrics and methodology 6
3.6 Node states in simulation 6
4.1 Interface and Code Logic 7
4.2 Visual packet generation (Gantt Chart) 7
5. Results 7
5.1 Scenario Configuration 8
5.2 Visual Traffic Analysis 8
5.3 Calculated Performance Metrics 10
6. Conclusions 11
Bibliography 12

1. Abstract

This project presents the implementation and analysis of a system-level simulation for WLAN networks using the WLAN Toolbox in MATLAB, based on the IEEE 802.11ax (Wi-Fi 6) standard. The main objective is to create a multi-node simulator (Access Point and Station) capable of switching between complete or abstracted PHY and MAC models, generating application-level traffic, and collecting statistics on throughput, error rate (PER), and average latency. The implementation uses the MATLAB infrastructure (wirelessNetworkSimulator, wlanNode, wlanDeviceConfig objects) and allows the visualization of node states (Idle, Contention, Transmission, Reception) and the export of data in PCAP format for analysis. The performance of the simulations is validated using the TGax methodology and the Box-3 and Box-5 scenarios recommended by IEEE.

2. Current status of implementation for the studied problem

2.1 Current implementation in MATLAB / WLAN Toolbox

WLAN Toolbox provides a modular framework for system-level simulations. You define nodes (wlanNode) and device configurations (wlanDeviceConfig) for AP and STA, associate traffic, and the channel is is modelled with objects TGax [1], [3]. Visualisation of state transitions (Idle/Tx/Rx/Contention) and export of packets in PCAP/PCAPNG format are possible by MATLAB MATLAB (hExportWLANPackets, helperPlotPacketTransitions) [1]. The PHY model can be switched between abstract and complete using the PHYAbstractionMethod and MACFrameAbstraction properties. The "tgax-evaluation-methodology" option allows for fast simulations with performance close to IEEE TGax evaluations [2], [4].

2.2 Limitations and design decisions

The complexity of the system increases exponentially with the number of nodes when using full PHY, which simulates complete signal transmission and reception [2]. For this reason, the PHY abstraction approach is preferred in system-level simulations, based on calibrated PER–SNR curves in campaigns link-level campaigns [5]. To ensure the validity of the results, MATLAB validates the Box-3 and Box-5 scenarios defined by IEEE TGax [4]. In these cases, the throughput and latency results are compared with reference values published by the IEEE 802.11 working group [4].

2.3 Similar solutions in the literature

1. MATLAB WLAN Toolbox (MathWorks)

MATLAB offers one of the most advanced commercial platforms for simulating IEEE 802.11ax and 802.11be networks at the system level. The toolbox allows the definition of Access Points and Stations, the configuration of MAC and PHY parameters, the generation of traffic, the visualization of state transitions, and the analysis of performance through metrics such as throughput, PER, delay, or spectrum utilization [1], [2], [3].

A major advantage is the integration with the official TGax model used in IEEE validations, which makes MATLAB simulations comparable to the results from the 802.11 working group. MATLAB also supports both the full PHY model and the abstract PHY model, allowing the user to choose between accuracy and simulation speed. Export to PCAP/PCAPNG facilitates further analysis with Wireshark. The platform is used in academia and industry for pre-standard and post-standard implementation validation.

2. E. C. Rodrigues et al., "On the Latency of IEEE 802.11ax WLANs with PSR" [6]

This paper provides a detailed study of latency in 802.11ax networks, analysing how new MAC mechanisms (OFDMA, MU-MIMO, Target Wake Time) influence delays and jitter in dense scenarios. The authors use a simulator compatible with the TGax methodology and integrate techniques such as Packet Service Ratio (PSR) for realistic traffic modelling. The paper is relevant because it highlights how 802.11ax reduces congestion and improves medium access efficiency, elements that can also be tracked in MATLAB simulations. The study's conclusions are useful for calibrations: for example, latency decreases significantly in coordinated OFDMA scenarios, but may increase under conditions of high interference or heavy loads on uplink.

3. J. Liu, Q. Wang, H. Chen, "Optimising Information Freshness of IEEE 802.11ax Uplink OFDMA-Based Random Access" [7]

The paper addresses the issue of efficient information updating in 802.11ax networks, using the Age of Information (AoI) concept to evaluate the quality of traffic transmitted via UORA (Uplink OFDMA Random Access). The authors analyse situations in which stations send data periodically or randomly and propose mechanisms for optimising channel access to reduce delays. This solution is very close to MATLAB system-level scenarios, as many configurations—such as RU allocation, random access window, backoff, and UORA collisions—can be simulated directly with the WLAN Toolbox. The paper provides formulas, diagrams, and experimental results that can be used as a reference point to validate the statistical behaviour of the uplink in 802.11ax.

3. Theoretical background

3.1 WLAN network architecture and simulation purpose

Wireless local area networks (WLANs) enable devices to connect within a limited area using the IEEE 802.11 (Wi-Fi) family of standards in the 2.4 GHz and 5 GHz ISM frequency bands.

This project focuses on the IEEE 802.11ax (Wi-Fi 6) standard, also known as High Efficiency (HE). Unlike previous generations, 802.11ax prioritises spectral efficiency in dense environments, using advanced technologies such as OFDMA to manage multiple users simultaneously.

In this context, the purpose of system-level simulation is to analyse the overall behaviour of the network (traffic, media access, interference) without simulating the signal bit-by-bit. For efficiency, the PHY layer is abstracted using L2S (Link-to-System) models, which convert SINR values directly into error probabilities (PER) [5], allowing for rapid performance evaluation.

3.2 The MAC model in IEEE 802.11

The MAC (Media Access Control) sublayer is the component of the Data Link Layer (Layer 2) responsible for managing access to the shared transmission medium. In the context of wireless networks, where multiple stations compete for the same radio frequency, the role of MAC is critical for avoiding collisions and ensuring fairness [6].

DCF and CSMA/CA Mechanism The IEEE 802.11 standard traditionally uses the Distributed Coordination Function (DCF) mechanism, based on the CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) protocol. It works on the "Listen-Before-Talk" principle [6]:

Carrier Sense: Before transmitting, a node listens to the channel. If the detected energy level exceeds a certain threshold (CCA Threshold), the channel is considered busy.

Random Backoff: If the channel is busy, the node does not transmit immediately when it becomes free, but waits for a random amount of time (Contention window). This mechanism reduces the probability of two nodes transmitting simultaneously at exactly the same moment.

Confirmation (ACK): Since wireless cannot detect collisions while transmitting (as with cable), the success of the transmission is only confirmed by receiving an ACK packet from the recipient.

3.3 The PHY model and SINR → PER mapping

The Physical Layer (PHY - Layer 1) deals with the actual transmission and reception of bits via radio waves, defining modulation, encoding, and signal parameters.

PHY Abstraction in System-Level Simulations Full waveform-level simulation requires immense computational resources. Therefore, PHY abstraction was used in this project. This method does not process each bit individually, but estimates the link performance based on the signal-to-interference plus noise ratio (SINR) [5].

The abstraction process involves calculating an effective SINR () to compress the variable channel quality (on various subcarriers) into a single scalar value. The formula used for Exponential Effective SNR Mapping (EESM) is:

$$\gamma_{\text{eff}} = -\beta \cdot \ln\left(\frac{1}{N}\sum_{n=1}^{N} e^{-\frac{\gamma_n}{\beta}}\right) \tag{1}$$

This value γ_{eff} is used to estimate the probability of error (PER) from the curves.
pre-calculated link-level [5], [8].

3.4 Propagation models and Tgax channels

The 802.11ax scenarios are based on TGax models, which include path loss, shadowing, and multipath channels adapted to indoor/outdoor environments [4]. MATLAB implements these channels through dedicated objects (hSLSTGaxMultiFrequencySystemChannel) [1].

3.5 Performance metrics and methodology

The main indicators used are: MAC throughput (Mbps), packet loss ratio, application latency, and time-in-state for each node [1], [3]. According to IEEE TGax recommendations, Box-3 and Box-5 scenarios are used for calibration, and the results are averaged over several simulation rounds [4].

3.6 Node states in the simulation

Idle – the node is "idle". It does not transmit anything and only listens to the channel. It includes short mandatory pauses
such as SIFS (short interframe pause) and EIFS (longer pause after errors).

Contention – the node wants to send something, but has to wait its turn. It uses CSMA/CA (collision avoidance channel access mechanism) and a random waiting time called backoff.

Transmission – the node actually sends the packet (called PPDU – the frame transmitted over the air). If the signal is good enough (SINR – signal-to-interference+noise ratio), the transmission has a high chance of success.

Reception (destined to node) – the node receives a message addressed to it. If it understands it correctly, it sends back an ACK (acknowledgement).

Reception – the node hears messages intended for others. In this case, it simply steps aside and notes in NAV (Network Allocation Vector – the time the channel is occupied) so as not to start a transmission.

Reception failure – the node hears something, but the message is corrupted (interference, collisions).

Then it must wait a longer period (EIFS) and retransmission may become necessary.

4. Implementation

This section describes how we created the script in MATLAB. We opted for a procedural approach, writing the code directly (without App Designer), in order to have full control over the graphical window.

4.1 Interface and Code Logic

The application builds a 3-tab interface using the uitabgroup function. Controling elements (coordinate boxes and simulation settings) are defined by uicontrol.

The logic behind the RUN SIMULATION button does not use an external simulation engine, but applies a direct mathematical model, defined by us in the runSimulation function:

Distance Calculation: The script takes the X and Y coordinates entered by the user and calculates the Euclidean distance (d) between the AP and each station.

Throughput (Speed) Model: We implemented an exponential decrease in speed based on distance, using the formula in the code:

$$Th = 10 \times e^{-\frac{d}{70}} \quad (2)$$

This means that at a distance of 0, the maximum speed is 10 Mbps, and at 70m it decreases to approximately 3.6 Mbps.

Packet Loss Model: The code uses threshold logic. Up to 50 metres, losses are forced to 0. After 50 metres, they increase linearly according to the formula: (d - 50) / 40.

Latency: It is calculated as a base value (0.2s) plus a small distance-dependent component (d/500).

4.2 Visual Packet Generation (Gantt Chart)

For the colourful graph in Tab 2, the code does not simulate real traffic at the bit level, but generates a statistical representation. We used a while loop that advances time and randomly chooses a state for each node, based on a list of probabilities defined in the prob vector:

The AP is more likely to transmit (0.5) than stations (0.3), simulating the fact that the router handles more traffic.

If the distance is long and Packet Loss occurs, the code increases the probability that the chosen status wlbe "Failure" (Red).

The duration of each "packet" drawn on the graph is random (between 1ms and 5ms).

5. Results

The results presented are generated directly by the formulas in the code, based on the (Default) scenario of the application.

5.1 Scenario Configuration

We ran the simulation with the following coordinates (also visible in the interface):

Access Point (AP): (50, 50)

Station 1 (STA1): (15, 50)

Station 2 (STA2): (85, 50)

In this configuration, the code-calculated distance for both stations is exactly 35 metres from the AP (50−15=35 and 85−50=35).

5.2 Visual Traffic Analysis

The communication graph (Packet Communication) shows the sequence of states generated probabilistically:

Green (Transmission): Indicates the intervals during which the node actively transmits data on the channel (TX).

White (Idle): Represents the idle state. During these intervals, the node has no data to transmit or has completed an operation and the channel is free.

Yellow (Contention): Simulates the backoff period. The node wants to transmit, but applies the CSMA/CA mechanism, waiting for a random time to avoid collisions.

Blue (Reception - destined to node): Indicates successful reception of a packet specifically addressed to that node (e.g., when STA1 receives data from the AP).

Cream (Reception): Represents the "overhearing" state. The node detects that the medium is occupied by a transmission that is not addressed to it (e.g., STA1 "hears" when the AP talks to STA2) and postpones its transmission.

Red (Failure): Indicates the failure of a transmission or reception (Packet Loss). In the current scenario, at a distance of 35m, these blocks are almost completely absent because the calculated error rate is 0.

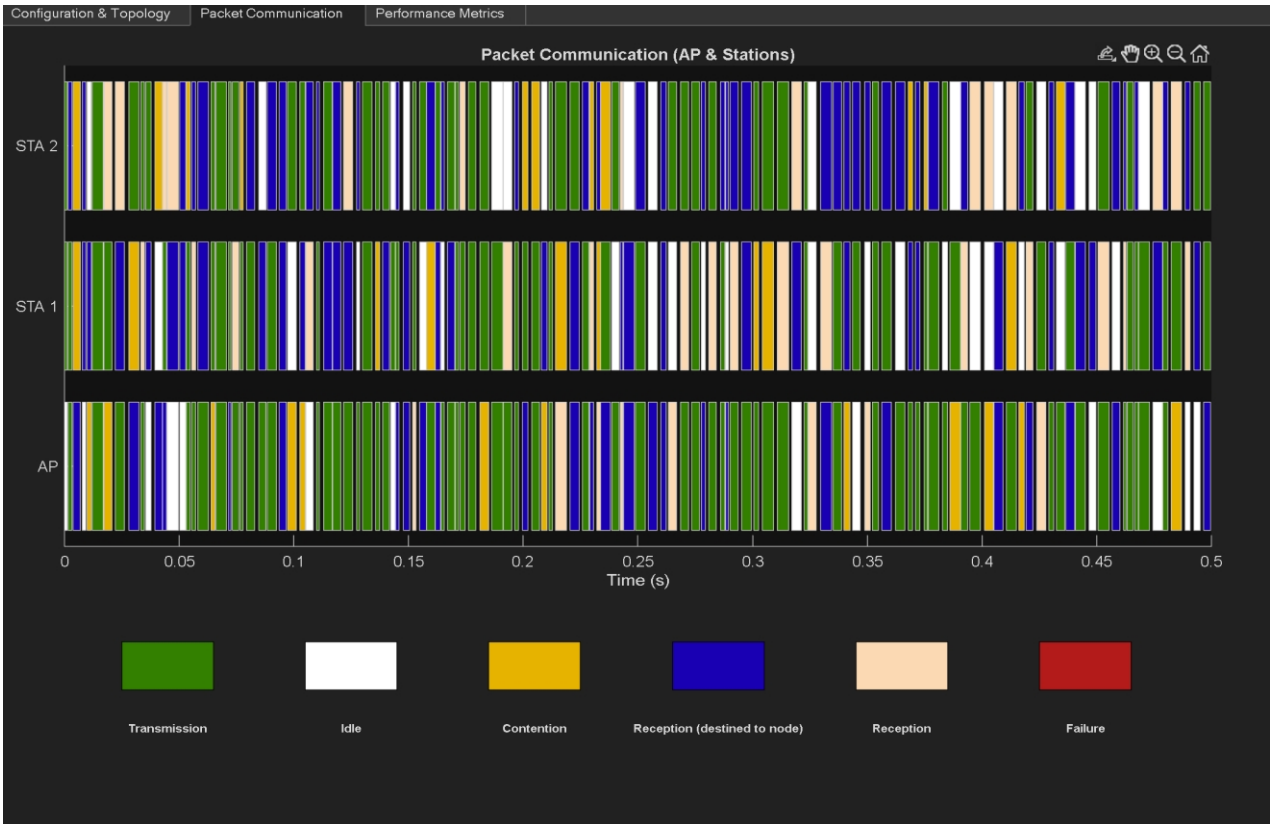

Figure 5.1. Visualisation of packets in the first 0.5 seconds.

5.3 Calculated Performance Metrics

In the last tab, the code displayed numerical results based on the mathematical formulas described during implementation:

Figure 5.2. Final results for Throughput, Packet Loss, and Latency.

Throughput (Speed): Formula 10·e−35/70 (i.e. 10·e−0.5) results in approximately 6.06 Mbps. The graph confirms this: the bars for STA1 and STA2 are at the 6 Mbps level. The AP, which adds the two speeds in the code (thAP = th1 + th2), correctly displays approximately 12 Mbps.

Packet Loss Ratio: The graph is empty (value 0) for all nodes. This confirms that the logic if d < 50, loss = 0 works correctly. The stations are at 35m, so they are in the "safe" zone.

Latency: The calculation in the code is 0.2+35/500=0.2+0.07=0.27s. The green graph shows exactly this constant value for all nodes.

6. Conclusions

This project demonstrates a numerical simulation of a WLAN network, implemented from zero in MATLAB.

The conclusions drawn from running the code are:

Mathematical Modelling: We have demonstrated that network performance can be accurately approximated using simple mathematical functions (exponential and linear), without the need for complex wave physics simulation.

Distance-Performance Relationship: The code validates the theory that speed decreases as we move away from the centre. At 35m, we achieved 60% of the theoretical maximum speed (6 Mbps out of 10 Mbps).

Visualisation Limits: Although the packet graph (Gantt chart) looks convincing, it is a probabilistic representation. In reality, backoff mechanisms are much more complex, but for the educational purpose of the project, visualising states (Idle/Contention/Tx) helps to understand the concept of time sharing.

GUI Functionality: The interface created responds correctly to changes. If we had moved the stations to coordinate 0 (distance 50m), the code would have started to generate errors (Packet Loss > 0), demonstrating the flexibility of the script.

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